

CHEMICAL KINETICS...

SUB-TOPICS

* Units: $1 \text{ Kilojoule} = 10^3 \text{ Joule}$

$= 8.3 \times 2.303 \Rightarrow R_T$

$\Rightarrow 19.11$

- Rate of reactions
- Order, General formula of K (Units)
- Graphs of reactions
- Law of Mass action
- Factors influencing rate of reaction
- Molecularity
- Half life.
- Formulas of Kinetics
- Arrhenius Equation
- Temperature coefficient
- Determining rate law exp based on Reaction Mechanism.

Zero Order.

1. $\frac{x_1}{t_1} = \frac{x_2}{t_2}$ 2. $t_{75\%} = \frac{3}{2} t_{50\%}$ 3. $t_{100\%} = 2 \times t_{50\%}$

* * First order.

$t_{75\%} = 2 t_{50\%}$

$t_{90\%} = \frac{10}{3} t_{50\%}$

$t_{87.5\%} = 3 t_{50\%}$

$t_{99\%} = 9 t_{90\%}$

$t_{93.75\%} = 4 t_{50\%}$

$t_{99.9\%} = 3 t_{90\%}$

$t_{99.9\%} = 10 t_{50\%}$

$t_{99.9\%} = 4 t_{90\%}$

FORMULAS

KEY CONCEPTS

- : Molecularity is Always a whole Number
- : For decomposition reactions $t_{50\%}$ is Constant
Hence, $t_{1/2} = 100 \text{ min}$ Order would be One.
- : Rate constant is independent of concentrations.
- : For First Order Rxn $t_{50\%}$ is Constant
- : Catalyst decreases Activation Energy Hence increase in rate of reaction.
- : rate of Exo reactions increases with increasing temp.
- : rate of Endo reactions increases with decreasing temp.

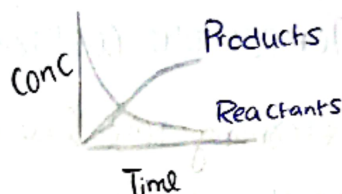
$$\text{Rate} = \frac{\text{conc}}{\text{Time}}$$

Law of Mass Action:

$$r \propto [A]^x [B]^y$$

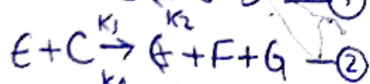
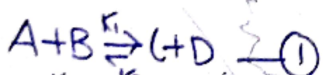
$$r = k[A]^x [B]^y$$

$$k = \frac{\text{Rate}}{[C]} \quad \text{Units: } \frac{\text{Mole/L}}{\frac{\text{sec}}{\text{Mole}}}$$



$$\text{Rate constant } k = \left(\frac{\text{Mole}}{\text{L}} \right)^{1-n} \text{sec}^{-1}$$

- : As rate Doubles rate constant also doubles
- : Rate Exp Should not be as intermediate!



$$\text{Rate} = k_3 [C][E]$$

$$K_{eq} = \frac{k_1}{k_2} = \frac{[C][D]}{[A][B]}$$

$$[C] = K_{eq} \frac{[A][B]}{[D]}$$

$$\text{Rate} = K_3 K_{eq} \frac{[A][B][C]}{[D]}$$

$$** \left(\frac{t'_{1/2}}{t''_{1/2}} \right) = \left(\frac{a''}{a'} \right)^{n-1}$$

$$t_{1/2} \propto \frac{1}{a^{n-1}}$$

$$K = \frac{2.303}{t} \log \frac{a}{a-x}$$

For First Order: Sec^{-1}

$$\bullet t_{1/2} = \frac{0.693}{K_1}, \quad t_{1/2} = \frac{0.693}{K_1 + K_2}$$

$$\bullet r = K_1 [A]^1$$

$$\bullet K = \frac{2.303}{t} \log \frac{P_i}{2P_i - P_f}$$

For Second Order: $\text{Mol}^{-1} \text{Sec}^{-1}$

$$r = K_2 [A]^2$$

$$t_{1/2} \propto \frac{1}{a}$$

$$t_{1/2} = \frac{1}{K a} \quad \boxed{K = \frac{1}{t} \left[\frac{1}{a-x} - \frac{1}{a} \right]}$$

$$\text{Zero Order: } t_{\text{soy.}} = \frac{a_0}{2K}$$

$\text{Mol l}^{-1} \text{s}^{-1}$

$$\text{Rate} = K[A]^0 \quad (\because K = \frac{c_0 - c}{t})$$

$$\text{Rate} = \frac{c_0 - c}{t}$$

* n^{th} Order Reaction:

$$t_{\text{soy.}} = \frac{1}{K(n-1)} \left[\frac{2^{n-1} - 1}{a^{n-1}} \right]$$

$$\text{Arrhenius Equation: } K = A e^{\frac{-E_a}{RT}}$$

• If $T \rightarrow \infty$ then $K = A$.

: For First Order time Period is infinite

$$\frac{1}{(t)_{50\%}} - \frac{1}{(t)_{50\%}} = \frac{x}{t}$$

$$K = \frac{2.303}{t} \log \frac{V_{\infty} - V_0}{V_{\infty} - V_t} //$$

Molecular Collision Theory:

- $\Delta G = -Ve$
- No. of Collision.
- Effective Collision.
- Energy factor.
- Orientation.

$$R = Z e^{-\frac{E_a}{RT}}$$

Energy factor
Collision factor.

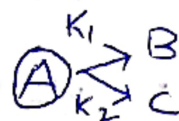
Transition State Theory:

$$\text{Rate} \propto \frac{1}{E_a}$$

$$\Delta H = E_{af} - E_{ab}$$

$$\text{Exo: } E_{af} < E_{ab}$$

$$\text{Endo: } E_{af} > E_{ab}$$



$$\% B = \frac{n_1 K_2}{n_1 K_1 + n_2 K_2} \times 100\%$$

$$\% C = \frac{n_2 K_2}{n_1 K_1 + n_2 K_2} \times 100$$

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

Q : Temp Coeff:

- Range 1.5 - 3.0.

$$\log \left(\frac{K_2}{K_1} \right) = \frac{E_a}{2.303 RT} \left[\frac{1}{T} - \frac{1}{T+10} \right]$$

$$\frac{K_2}{K_1} = e^{\frac{\Delta T}{10}} \quad [Q \approx 2]$$

$$K_{eq} = e^{-\frac{\Delta H}{RT}}$$

$$\propto \frac{1}{2} \propto (P_0)^{1-n}$$

where

$P_0 \rightarrow$ Pressure

$n \rightarrow$ Order...

: Slope increases Order increases 1

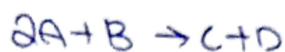
Positive Catalyst:

Negative Catalyst:

Promoter: Catalyst of Catalyst:

Order cannot be determined by Theoretically

For



Rate = $k[A]^x[B]^y$ $x+y \rightarrow$ Order:

Types of Reaction:

1. Elementary Reaction.
2. Complex Reaction.

Elementary Reaction:

The reactions which complete in single step is called

Complex reactions:

The reactions which complete in more than single step is called

Slow step is also called as Rate determining step.

Molecularity:

- The no. of molecules which participate in the reaction.
- Molecularity always +ve not in -ve fraction and not equal to 0.

Max Molecularity : 3

Min Molecularity : 1

Can be calculated by Elementary reactions.

Order :

- Sum of the Powers of Concentrations of a Reactants When it is expressed in Rate Equation.
- Order May be 0 or fraction.
- in Elementary reactions coeff can be considered as Equal to Order :

$$\text{Rate} \propto [A][B]$$

$$\text{Rate} \propto [A]^n$$

$$\text{Rate} = k[A]^n$$

$$k = \frac{\text{Rate}}{[A]^n}$$

Units of $k = \frac{\text{Rate}}{[A]^n} = \frac{(\frac{\text{Mole}}{\text{L}})}{\text{Sec}(\frac{\text{Mole}}{\text{L}})^n}$

$$\boxed{(\text{Mole/L})^{1-n} \text{Sec}^{-1}}$$

$$n=0 : \text{Mole L}^{-1} \text{Sec}^{-1}$$

$$n=1 : \text{Sec}^{-1}$$

$$n=2 : \text{L} \cdot \text{Mole}^{-2} \text{Sec}^{-1}$$

$$n=3 : \text{L}^2 \text{Mole}^{-3} \text{Sec}^{-1}$$

$$n=-1 : \text{Mole}^2 \text{L}^{-2} \text{Sec}^{-1}$$

: Rate constant is independent of concentrations |

Second Order:

$$K = \frac{1}{t} \left[\frac{1}{A_t} - \frac{1}{A_0} \right]$$

$$t_{1/2} = \frac{1}{K} \times \frac{1}{a_0} \Rightarrow \boxed{t_{1/2} = \frac{1}{Ka_0}}$$

$$K = \frac{2.303}{t} \log \frac{a}{a-x}$$

$$t_{50\%} = \frac{2.303}{K} \log \frac{100}{50}$$

$$t_{50\%} = \frac{0.693}{K}$$

Equal drops in Equal time interval:

$$\frac{2.303}{t_{50}} \log \frac{100}{50} = \frac{2.303}{t_{75}} \log \frac{100}{25}$$

$$\frac{t_{75}}{t_{50}} = 2$$

$$t_{75} = 2 \times t_{50}$$

$$t_{99.9} = 10 t_{50}$$

Average life:

$$\boxed{\lambda = \frac{1}{K}}$$

$$a \rightarrow \frac{a}{2} \rightarrow \frac{a}{4} \rightarrow \frac{a}{8} \rightarrow \frac{a}{2^n}$$

After n -half lives left amount: $\frac{a_0}{2^n}$

reacted: $a_0 - \frac{a_0}{2^n}$

: for first Order Rxn $t_{50\%}$ is constant ... //